## Line counts (scc)

**Generated by `just line-counts`; do not edit by hand.** `tests.fs` is excluded from the per-module table and reported separately.

**Implementation (F#):** 130 files, 53428 lines, 42761 code. `tests.fs`: 14432 lines, 11924 code. **Kernel:** 17 files, 3584 lines, 2766 code. **Prelude core (excluding tests, tools, demo/acceptance, and edge/verification):** 64 files, 21442 lines, 17324 code. **Edge and verification (not part of the initial port):** 22 files, 5884 lines, 4473 code. **Demo and acceptance (not counted as Prelude code):** 10 files, 1565 lines, 1131 code. **Test harness (tests, granular, properties):** 5 files, 17903 lines, 14652 code. **Tools and local scripts:** 12 files, 3050 lines, 2415 code. The categories partition the implementation total: kernel, Prelude core, edge/verification, demo/acceptance, test harness, and tools. The table below excludes `tests.fs` and demo/acceptance: 119 files, 37431 lines, 29706 code.

**Documentation:** 93 files, 29353 lines, 19036 code.

| Module | Lines | Code | Complexity |
|---|---|---|---|
| components/verification/tests/properties.fs | 1834 | 1379 | 147 |
| components/prelude/evaluator/dispatch.fs | 1500 | 1227 | 171 |
| components/verification/tests/granular-tests.fs | 1389 | 1152 | 87 |
| components/prelude/scheduler.fs | 1417 | 1095 | 106 |
| components/core/prose-parser.fs | 1107 | 961 | 144 |
| components/kernel/prelude-image.fs | 1015 | 915 | 96 |
| components/book-host/book-runner.fs | 992 | 840 | 69 |
| components/prelude/evaluator/value-sets.fs | 884 | 680 | 98 |
| components/prelude/evaluator/environment.fs | 758 | 613 | 110 |
| components/verification/typing/smt.fs | 819 | 608 | 68 |
| components/core/runtime-types.fs | 820 | 604 | 6 |
| components/prelude/evaluator/class-system.fs | 731 | 603 | 80 |
| components/core/pattern-portable.fs | 678 | 590 | 85 |
| components/prelude/evaluator/lowering.fs | 640 | 572 | 27 |
| bases/scale-cli/entry.fs | 695 | 554 | 68 |
| components/core/fsharp-abstraction.fs | 611 | 537 | 59 |
| components/verification/typing/type-retrieval.fs | 602 | 537 | 29 |
| components/prelude/evaluator/cell-system.fs | 641 | 536 | 70 |
| components/core/expression-portable.fs | 558 | 520 | 59 |
| components/verification/typing/type-lattice.fs | 679 | 520 | 51 |
| components/kernel/abi.fs | 628 | 510 | 68 |
| components/prelude/evaluator/pattern.fs | 555 | 482 | 72 |
| components/core/package-portable.fs | 540 | 477 | 45 |
| components/prelude/evaluator/channel-event.fs | 569 | 474 | 68 |
| components/prelude/evaluator/scheduler.fs | 603 | 472 | 77 |
| components/verification/typing/violations.fs | 525 | 453 | 10 |
| components/prelude/evaluator/evaluator.fs | 542 | 438 | 51 |
| components/verification/typing/emitter.fs | 575 | 411 | 46 |

| Module | Lines | Code | Complexity |
|---|---|---|---|
| tools/architecture-check.fsx | 436 | 373 | 29 |
| tools/consistency-check.fsx | 498 | 363 | 49 |
| components/ipc/actor-host.fs | 430 | 360 | 51 |
| components/core/parser.fs | 432 | 355 | 58 |
| tools/fsharp-classifier.fsx | 385 | 323 | 38 |
| components/kernel/lowered-abstraction.fs | 547 | 313 | 67 |
| components/prelude/writer/il-writer.fs | 349 | 305 | 19 |
| components/kernel/term-portable.fs | 319 | 289 | 41 |
| components/prelude/evaluator/control.fs | 324 | 258 | 39 |
| components/ipc/wire.fs | 296 | 242 | 44 |
| components/kernel/prelude-execution.fs | 279 | 242 | 27 |
| components/verification/formal/refiner.fs | 340 | 241 | 23 |
| components/shell-host/shell-host.fs | 287 | 237 | 35 |
| components/core/vocabulary.fs | 231 | 229 | 0 |
| tools/update-test-counts.fsx | 310 | 225 | 16 |
| components/core/image-types.fs | 389 | 220 | 16 |
| components/prelude/evaluator/word.fs | 260 | 217 | 36 |
| tools/update-line-counts.fsx | 263 | 210 | 17 |
| tools/rename-names.fsx | 279 | 207 | 17 |
| components/core/persistence.fs | 304 | 204 | 30 |
| components/core/reclaim.fs | 246 | 194 | 16 |
| components/verification/formal/decision-table.fs | 273 | 183 | 13 |
| components/prelude/evaluator/condition.fs | 225 | 181 | 27 |
| tools/polylith-check.fsx | 225 | 181 | 33 |
| components/core/scale-file.fs | 254 | 178 | 22 |
| components/verification/formal/decision-table-surface.fs | 215 | 177 | 19 |
| components/prelude/evaluator/channel.fs | 264 | 175 | 20 |
| components/prelude/evaluator/signatures.fs | 191 | 172 | 1 |
| components/core/ownership-types.fs | 187 | 153 | 15 |
| components/prelude/evaluator/history.fs | 221 | 153 | 21 |
| components/prelude/evaluator/set-algebra.fs | 177 | 143 | 18 |
| components/verification/typing/domain-expression.fs | 186 | 132 | 18 |
| components/prelude/evaluator/ownership.fs | 184 | 128 | 17 |
| components/core/vocabulary-in-image.fs | 142 | 125 | 13 |
| components/verification/tests/differential.fs | 148 | 117 | 13 |
| components/prelude/evaluator/runtime-guard.fs | 157 | 115 | 25 |
| components/core/pattern-elaboration.fs | 129 | 112 | 17 |
| components/ipc/classification.fs | 124 | 108 | 14 |
| components/kernel/term-verification.fs | 138 | 105 | 8 |

| Module | Lines | Code | Complexity |
|---|---|---|---|
| components/kernel/store.fs | 128 | 101 | 11 |
| components/verification/tests/benchmarks.fs | 133 | 101 | 8 |
| components/prelude/evaluator/common.fs | 153 | 100 | 28 |
| components/core/expression-logic.fs | 128 | 97 | 5 |
| components/verification/typing/nearest-sentence.fs | 140 | 97 | 6 |
| components/verification/formal/discourse.fs | 136 | 93 | 15 |
| components/ipc/language-transport.fs | 106 | 90 | 13 |
| components/kernel/primitive-wire.fs | 117 | 88 | 13 |
| components/prelude/evaluator/quotation-run.fs | 107 | 85 | 7 |
| components/core/pattern-analysis.fs | 114 | 84 | 15 |
| components/core/epoch.fs | 119 | 83 | 7 |
| components/verification/tests/test-runner.fs | 100 | 80 | 7 |
| tools/module-dependencies.fsx | 105 | 80 | 17 |
| components/shell-host/prose-shell.fs | 110 | 79 | 7 |
| tools/decision-index.fsx | 103 | 79 | 9 |
| components/kernel/lowered-instructions.fs | 99 | 72 | 2 |
| components/cli/cli.fs | 92 | 71 | 9 |
| components/prelude/writer-runtime.fs | 95 | 71 | 9 |
| components/prelude/evaluator/program.fs | 107 | 70 | 8 |
| tools/freshness-check.fsx | 96 | 68 | 9 |
| components/verification/typing/presentation.fs | 97 | 67 | 12 |
| components/core/tokenizer.fs | 78 | 64 | 10 |
| components/prelude/evaluator/closure.fs | 97 | 63 | 7 |
| components/core/pattern-logic.fs | 85 | 60 | 0 |
| components/core/arena.fs | 78 | 58 | 2 |
| components/book-host/project-index.fs | 78 | 57 | 3 |
| components/core/package-logic.fs | 76 | 55 | 0 |
| components/prelude/evaluator/bookmark.fs | 82 | 53 | 5 |
| components/book-host/book-host.fs | 64 | 50 | 10 |
| components/prelude/evaluator/with-fields.fs | 64 | 49 | 5 |
| components/prelude/evaluator/code.fs | 66 | 46 | 7 |
| components/prelude/evaluator/stack.fs | 59 | 45 | 7 |
| components/core/channel-store.fs | 65 | 43 | 8 |
| components/core/portable-json.fs | 45 | 31 | 6 |
| components/verification/formal/decision-table-emitter.fs | 52 | 31 | 0 |
| components/core/vocabulary-list.fs | 35 | 28 | 7 |
| components/verification/formal/diagnostics.fs | 47 | 25 | 1 |
| components/core/host-words.fs | 35 | 23 | 7 |
| components/kernel/kernel-state.fs | 36 | 23 | 0 |

| Module | Lines | Code | Complexity |
|---|---|---|---|
| components/kernel/types.fs | 41 | 22 | 0 |
| components/kernel/symbolic-abstraction.fs | 27 | 21 | 0 |
| components/kernel/lexicon.fs | 25 | 17 | 0 |
| components/kernel/stack.fs | 26 | 17 | 3 |
| components/kernel/index-transformation.fs | 22 | 16 | 3 |
| components/prelude/evaluator/kernel-bridge.fs | 28 | 15 | 0 |
| components/kernel/scalar-read.fs | 18 | 11 | 1 |
| bases/scale-cli/demo/generated/ PlayerSpecificationPristine.fs | 16 | 5 | 2 |
| bases/scale-cli/demo/generated/ PlayerSpecificationRefined.fs | 16 | 5 | 3 |
| components/kernel/tensor.fs | 119 | 4 | 0 |
| components/nu-host/nu-host.fs | 10 | 4 | 0 |
| projects/scale-tests/entry.fs | 4 | 3 | 0 |
| .SageFs/config.fsx | 1 | 1 | 0 |